Chapter 1

$$\begin{cases} 1C = 1A \times 1sec \\ 1e = 1.602\times10^{-19}C \\ 1C = 6.24\times10^{18}e \end{cases}$$

$$i = \frac{dq}{dt}$$

$$q(t) = \int_{t_0}^{t} i\,dt + q(0)$$

$$V = \frac{dw}{dq}$$

$$p = \frac{dw}{dt} = vi$$

$$w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} vi\,dt$$

Chapter 2

§2.2 Ohm's Law $v = iR$

$G = \frac{1}{R}$ (S)

$$\begin{cases} P = vi = i^2R = \frac{v^2}{R} \\ P = vi = \frac{i^2}{G} = v^2G \end{cases}$$

§2.3

network - a circuit

branch, node, loop

mesh - loop that does not enclose any other loops.

$b = m + n - 1$

§2.4 Kirchhoff's Laws

KCL $\Sigma i = 0$

KVL $\Sigma v = 0$.

§2.5

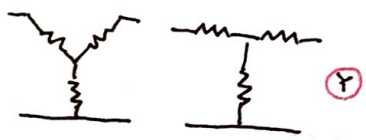
series: $R = \Sigma R_i$

principle of voltage divider: $v_i = \frac{R_i}{R} v$

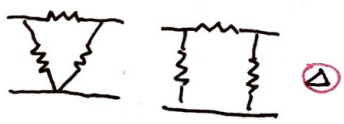
parallel: $G = \Sigma G_i$ ($\frac{1}{R} = \Sigma \frac{1}{R_i}$)

principle of current divider: $i_i = \frac{G_i}{G} i$

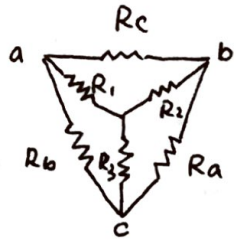
§2.7 Wye-Delta.



Y



Δ



Δ→Y

$$\begin{cases} R_1 = \frac{R_bR_c}{R_a+R_b+R_c} \\ R_2 = \frac{R_aR_c}{R_a+R_b+R_c} \\ R_3 = \frac{R_bR_c}{R_a+R_b+R_c} \end{cases}$$

Y→Δ

$$\begin{cases} R_a = \frac{R_1R_2+R_2R_3+R_1R_3}{R_1} \\ R_b = \frac{R_1R_2+R_2R_3+R_1R_3}{R_2} \\ R_c = \frac{R_1R_2+R_2R_3+R_1R_3}{R_3} \end{cases}$$

证明 对Y. 用KVL. R.
对Δ 用KCL. G
化出 v.

Chapter 3

§3.2 Nodal Analysis

① Select reference

② Assign $v_1, v_2, \dots, v_{n-1}$

③ KCL ④ Solve.

Nodal Analysis by Inspection

$$\begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1N} \\ G_{21} & G_{22} & \cdots & G_{2N} \\ \vdots & \vdots & & \vdots \\ G_{N1} & G_{N2} & \cdots & G_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} = \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix}$$

G_{kk} - sum of conductances connected to node k.

$G_{kj} = G_{jk}$ - negative of the sum of the conductances directly connecting nodes k & j ($k \neq j$)

i_k - sum of current sources entering node k.

*Inspection is only used in cases without corresponding sources.

§3.3 Nodal Analysis with Voltage Source

① reference ⊖ nonreference

② nonreference ⊖ noreference

⇒ Supernode (with elements in parallel)

§3.4 Mesh Analysis

① Assign mesh currents $i_1, i_2, \dots, i_n$.

② KVL ③ Solve.

Mesh Analysis by Inspection

$$\begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1N} \\ R_{21} & R_{22} & \cdots & R_{2N} \\ \vdots & \vdots & & \vdots \\ R_{N1} & R_{N2} & \cdots & R_{NN} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix}$$

R_{kk} - sum of resistances in mesh k.

$R_{kj} = R_{jk}$ - negative of the sum of the resistances in common with meshes k & j ($k \neq j$).

V_k - sum taken clockwise of voltage sources in mesh k. with voltage rise treated as positive

§3.5 Mesh Analysis with Current Source

① ⊖ excusive

② ⊖ shared.

⇒ Supermesh (exclude elements in series with ⊖)

Nodal	Mesh
more parallel connected element	more series connected elements

mesh analysis

* planar

# of meshes = # of independent loops.

* 注意用大小写区分 current 和 mesh current

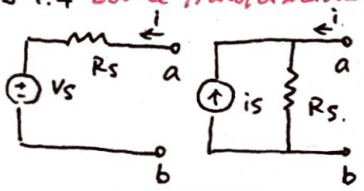
VE 215 Mid 1

Chapter 4

§4.2.4.3 Linearity & Superposition

* replace voltage sources with short cuts
* replace current sources with open circuits
* leave dependent sources.

§4.4 Source Transformation



$$\boxed{V_s = i_s R \text{ or } i_s = \frac{V_s}{R}}$$

§4.5 Thevenin's Theorem

V_{Th} - open-circuit voltage

R_{Th} - equivalent resistance when independent sources are turned off.

① No independent sources

$R_{Th} = R_{eq}$

② Independent sources.

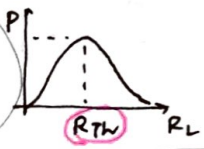
$R_{Th} = \frac{V_o}{i_o}$

§4.6 Norton's Theorem.

$R_N = R_{Th}$. $I_N = \frac{V_{Th}}{R_{Th}}$

§4.8 Maximum Power Transfer

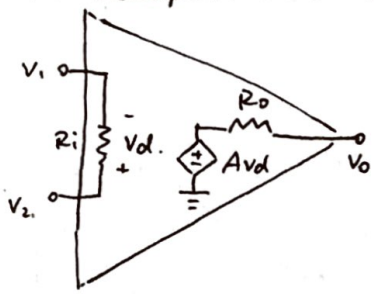
$P = i^2 R_L = \left(\frac{V_{Th}}{R_{Th}+R_L}\right)^2 R_L$



Chapter 5

§5.3 Ideal Operational Amplifier

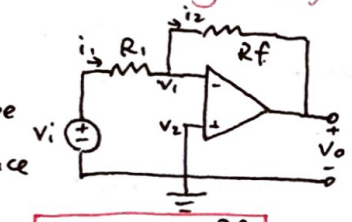
$A = \infty$ (open-loop gain)
$R_i = \infty$ (input resistance)
$R_o = 0$ (output resistance)



⇒ ① $i_1 = i_2 = 0$

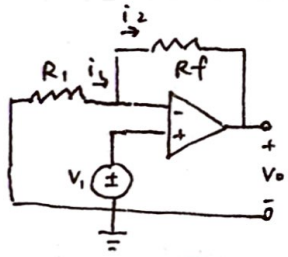
② $V_d = 0$

§5.4 Inverting Amplifier



$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{R_f}{R_1}}$$

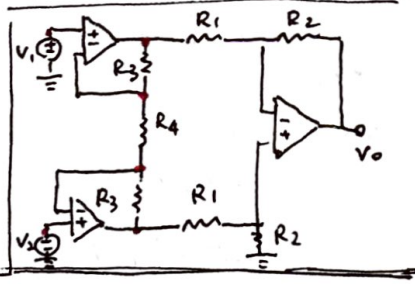
§5.5 Non-inverting Amplifier



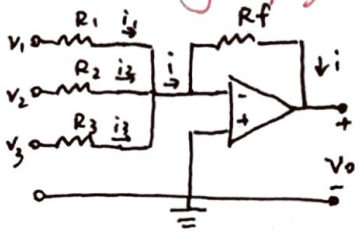
$$\boxed{A_v = \frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}}$$

$R_f = 0$ or $R_1 = \infty$. then $A_v = 1$.

voltage follower

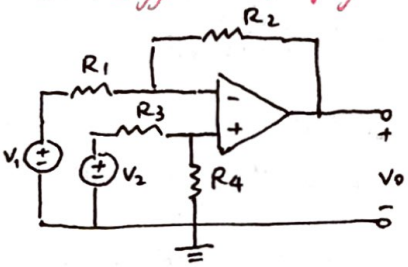


§5.6 Summing Amplifier



$$\boxed{V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)}$$

§5.7 Difference Amplifier

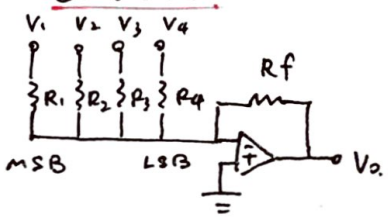


$$\boxed{V_o = \frac{R_1+R_2}{R_1}\,\frac{R_4}{R_3+R_4}V_2 - \frac{R_2}{R_1}V_1}$$

§5.8. Cascaded Op Amp Circuits

Stage.

§5.10.1 Digital-to-Analog Converter.



$$-V_o = \frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3 + \frac{R_f}{R_4}V_4$$

$$= k(2^3V_1 + 2^2V_2 + 2^1V_3 + 2^0V_4)$$

§5.10.2. Instrumentation Amplifiers.

$V_o = \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_4}\right)(V_2 - V_1)$

$R_4 = RG$. → can be used to adjust A.

$R_3 = R_2 = R_1 = R$, then

$V_o = A_v(V_2 - V_1) = A_v V_d$.

$V_d = V_2 - V_1$

$V_c = \frac{V_1 + V_2}{2}$.

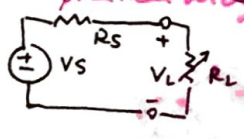
⇔ $V_2 = \frac{1}{2}V_d + V_c$

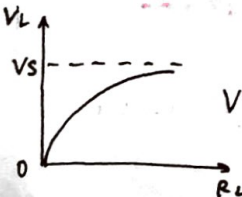
$V_1 = -\frac{1}{2}V_d + V_c$.

* input resistance is high

Source modeling (§4.10)

practical voltage source





$V_L = \frac{R_L}{R_s + R_L}V_s$.

① $R_s \ll R_L$ ideal

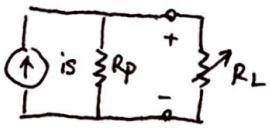
② open-circuited $V_{oc} = V_s$.

unloaded

with R_L. $V_L \downarrow$.

loading effect

practical current source



$i_L = \frac{R_p}{R_p + R_L}i_s$

① $R_p \gg R_L$ ideal

② short-circuited $i_{sc} = i_s$

unloaded

with R_L, $i_L \downarrow$

loading effect

§6 Capacitors & Inductors

(def) $q = Cv$

$i = C\frac{dv}{dt}$

$v = \frac{1}{C}\int_{t_0}^{t} i\,dt + v(t_0)$

$w = \frac{1}{2}Cv^2$

series $C_{eq} = \sum C_i$

parallel $\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$

(def) $\psi = Li$

$v = L\frac{di}{dt}$

$i = \frac{1}{L}\int_{t_0}^{t} v\,dt + i(t_0)$

$w = \frac{1}{2}Li^2$

series $L_{eq} = \sum L_i$

parallel $\frac{1}{L_{eq}} = \sum \frac{1}{L_i}$

§7 First-Order Circuits

RC Circuits

$\tau = RC$

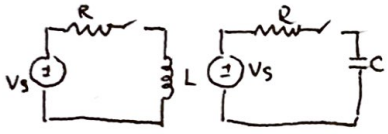
$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-\frac{t}{\tau}}$

$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}}$

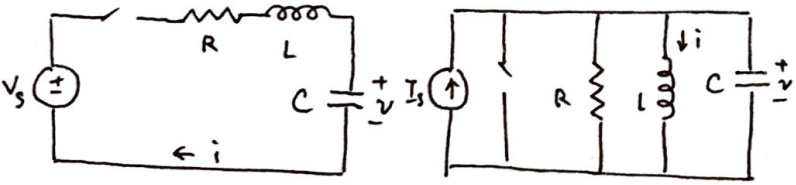
RL Circuits

$\tau = \frac{L}{R}$ (equivalent resistence at L)

$*\ e^{-\frac{5\tau}{5}} \approx 0$



§8 Second-Order Circuits



Series RLC Circuit

$\begin{cases} iR + L\frac{di}{dt} + v = 0\ (V_s) \\ i = C\frac{dv}{dt} \end{cases}$

$\Rightarrow s^2 + \frac{R}{L} + \frac{1}{LC} = 0\ (\frac{V_s}{LC})$

$\alpha = \frac{R}{2L}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{1}{LC}v = \frac{1}{LC}v_c$

Parallel RLC Circuit

$\begin{cases} \frac{v}{R} + i + C\frac{dv}{dt} = 0\ (I_s) \\ v = L\frac{di}{dt} \end{cases}$

$\Rightarrow s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0\ (\frac{I_s}{LC})$

$\alpha = \frac{1}{2RC}$ $\omega_0 = \frac{1}{\sqrt{LC}}$

$\frac{d^2i}{dt^2} + \frac{1}{RC}\frac{di}{dt} + \frac{1}{LC}i = \frac{1}{LC}I_s$

① Over-damping $\alpha > \omega_0$

$x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

② Critical damping $\alpha = \omega_0$

$x(t) = (B_1 + B_2 t)e^{st}$

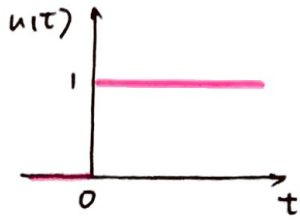
③ Under-damping $\alpha < \omega_0$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

$x(t) = e^{-\alpha t}(C_1\cos(\omega_d t) + C_2\sin(\omega_d t))$

§7.4 Singularity Functions

① Unit Step

$u(t) = \begin{cases} 0, & t<0 \\ 1, & t>0 \end{cases}$

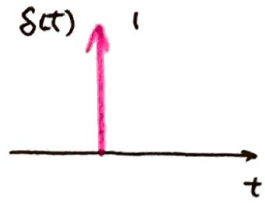


② Unit Impulse

$\delta(t) = \frac{d}{dt}u(t)$

$u(t) = \int_{-\infty}^{t}\delta(t)\,dt$

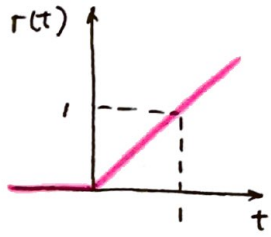
$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\,dt = f(t_0)$



③ Unit Ramp

$r(t) = \begin{cases} 0, & t \le 0 \\ t, & t \ge 0 \end{cases}$

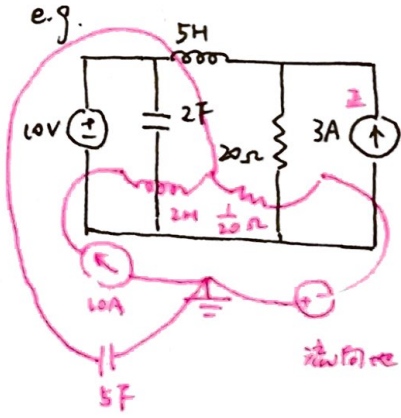
$r(t) = \int_{-\infty}^{t} u(\tau)\,d\tau = tu(t)$



§8.10 Duality

R	G
L	C
v	i
±	↑
Node	Mesh
Series	Parallel
Open	Short
KVL	KCL
Thevinin	Norton

e.g.



§11 AC Power Analysis

$p(t) = v(t)i(t) = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m\cos(2\omega t + \theta_v + \theta_i)$

$P = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) = \frac{1}{2}\mathrm{Re}(\tilde{V}\tilde{I}^*)$

$= \frac{1}{2}I^2R$

If $\theta_v = \theta_i$

$P = \frac{1}{2}V_m I_m = \frac{1}{2}I_m^2 R = \frac{V_m^2}{2R}$

If $\theta_v - \theta_i = \pm 90°$

$P = 0$

§9 Sinusoids & Phasors

sinusoid : $v(t) = V_m \cos(\omega t + \varphi)$

phasor : $\tilde{V} = V_m e^{j\varphi} = V_m \angle \varphi$

sinor : $\tilde{V} e^{j\omega t} = V_m e^{j(\omega t + \varphi)}$

$\Rightarrow v(t) = Re(\tilde{V} e^{j\omega t})$

Element	Time domain	Frequency domain
R	$v = Ri$	$\tilde{V} = R\tilde{I}$
L	$v = L\frac{di}{dt}$	$\tilde{V} = j\omega L \tilde{I}$
C	$i = C\frac{dv}{dt}$	$\tilde{V} = \frac{1}{j\omega C}\tilde{I}$

§11.3 Maximum Average Power Transfer

$Z_L = Z_{Th}^*$ $\quad P_{max} = \frac{V_{Th}^2}{8R_{Th}}$

§11.4 Effective / RMS Value

$P = I_{eff}^2 R \Rightarrow \begin{cases} I_{eff} = \sqrt{\frac{1}{T}\int_0^T i^2 dt} = I_{rms} \overset{\text{sinusoid}}{=} \frac{I}{\sqrt{2}} \\ V_{eff} = \sqrt{\frac{1}{T}\int_0^T v^2 dt} = V_{rms} = \frac{V}{\sqrt{2}} \end{cases}$

§10 Steady State Analysis

KCL KVL

Δ - Y conversion

Nodal Analysis

Mesh Analysis

Thevinin's Theorem

Norton's Theorem

Superposition

Source Transformation

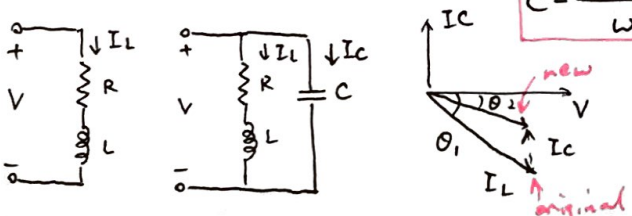
$S = \tilde{V}_{rms}\tilde{I}_{rms}^* = \frac{V_{rms}^2}{Z^*} = I_{rms}^2 Z$

$P = I_{rms}^2 R \quad Q = I_{rms}^2 X \quad (Z = R + jX)$

$= \frac{V_{rms}^2}{R} \quad = \frac{V_{rms}^2}{X}$

§11.8 Power Factor Correction

$$\boxed{C = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V^2}}$$



§9.5 Impedance & Admittance

$Z = \frac{\tilde{V}}{\tilde{I}}\ (\Omega) \qquad Y = \frac{\tilde{I}}{\tilde{V}}\ (S)$

$= R + jX \qquad = G + jB$

resistance reactance conductance susceptance

$X > 0$ inductive / lagging (i lags v)

$X < 0$ capacitive / leading (i leads v)

§11.5 Apparent Power & Power Factor

§11.6 Complex Power

$S = \tilde{V}_{rms}\tilde{I}_{rms} = |S| \angle(\theta_v - \theta_i)$

VA

$= V_{rms} I_{rms} \cos(\theta_v - \theta_i) + j V_{rms} I_{rms} \sin(\theta_v - \theta_i)$

$= P + jQ$

W VAR

power factor angle

$pf = \cos(\theta_v - \theta_i)$

① $Q = 0$ resistive load. unity pf

② $Q < 0$ capacitive ~ leading pf

③ $Q > 0$ inductive ~ lagging pf

* $e^{i\theta} = \cos\theta + i\sin\theta = \angle\theta$

$x + iy = \sqrt{x^2+y^2}\, e^{i\arctan\frac{y}{x}} = \sqrt{x^2+y^2}\angle\arctan\frac{y}{x}$

* $\frac{x\angle\alpha}{y\angle\beta} = \frac{x}{y}\angle(\alpha - \beta)$

$(x\angle\alpha)(y\angle\beta) = xy\angle(\alpha+\beta)$

* +, − rectangular form

×, ÷ angular form

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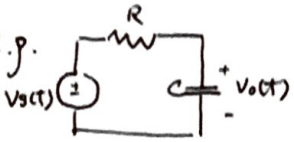
§14 Frequency Response

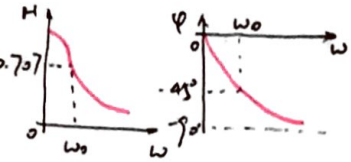
$H(\omega) = \frac{Y(\omega)}{X(\omega)}$ (transfer function = $\frac{\text{output phasor}}{\text{input phasor}}$)

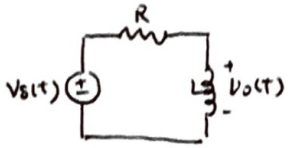
e.g.



$H(\omega) = \frac{1}{1+j\omega RC}$

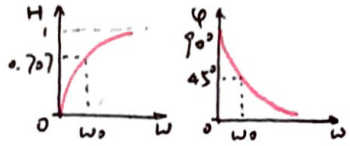
$\Rightarrow \begin{cases} H = \frac{1}{\sqrt{1+(\omega/\omega_0)^2}} \\ \varphi = -\arctan\frac{\omega}{\omega_0} \end{cases}$





$H(\omega) = \frac{j\omega L}{R+j\omega L}$

$\Rightarrow \begin{cases} H = \frac{\omega}{\sqrt{(1/\omega_0)^2+\omega^2}} \\ \varphi = \arctan\frac{1}{\omega_0\omega} \end{cases}$



Decibel Scale

power Gain $= \log_{10}\frac{P_2}{P_1}$

$G_{dB} = 10\log_{10}\frac{P_2}{P_1}$

$R_1 = R_2 \Rightarrow G_{dB} = 20\log_{10}\frac{V_2}{V_1}$

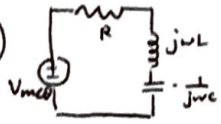
$= 20\log_{10}\frac{I_2}{I_1}$

$\bar{H} = H\angle\phi = He^{j\phi}$

$\ln\bar{H} = \ln H + j\phi$

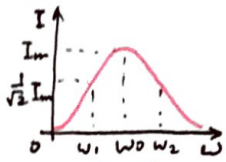
$H_{dB} = 20\log_{10}H$

Series Resonance



resonance $\Rightarrow Z = R + j\omega L + \frac{1}{j\omega C} = 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$

- * LC short circuit
- * V, I in phase
- * $I = I_{max}$



bandwidth B

$\boxed{\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}}$

$\omega_1\omega_2 = \omega_0^2$

$\boxed{BW = \omega_2 - \omega_1 = \frac{R}{L}}$

$\boxed{Q = \frac{\omega_0}{BW}}$

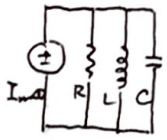
For high-Q circuit ($Q \geq 10$)

$\omega_{1,2} = \mp\frac{\omega_0}{2Q} + \omega_0 = \mp\frac{BW}{2} + \omega_0$

$\rightarrow Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$ $\quad BW = \frac{R}{L} = \frac{\omega_0}{Q}$

$V_L = V_C = QV_m$.

Parallel Resonance



resonance $\Rightarrow Y = \frac{1}{R} + j\omega C + \frac{1}{j\omega L} = 0 \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

- * LC open circuit
- * V, I in phase
- * $V = V_{max}$

$\boxed{\omega_{1,2} = \mp\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}}$

$\boxed{BW = \omega_2 - \omega_1 = \frac{1}{RC}}$ $\boxed{Q = \frac{\omega_0}{BW} = \omega_0 RC = \frac{R}{\omega_0 L}}$

Bode Plots

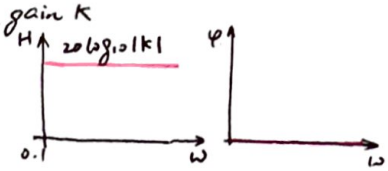
standard form

$$H(\omega) = \frac{K(j\omega)^{\pm 1}(1+j\omega/z_1)[1+j2\zeta_1\omega/\omega_k + (j\omega/\omega_k)^2]\cdots}{(1+j\omega/p_1)[1+j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]\cdots}$$

① constant term, the gain K

$H_{dB} = 20\log_{10}|K|$

$\varphi = \begin{cases} 0°, & K>0 \\ 180°, & K<0 \end{cases}$

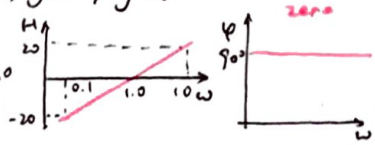


② pole/zero at the origin, $(j\omega)^{-1}$/$(j\omega)^1$

$(j\omega)$: $H_{dB} = 20\log_{10}\omega$, $\varphi = 90°$

$(j\omega)^{-1}$: $H_{dB} = -20\log_{10}\omega$, $\varphi = -90°$

$(j\omega)^N$: $H_{dB} = N\log_{10}\omega$, $\varphi = 90N°$



slope = 20dB/decade. (1~10)

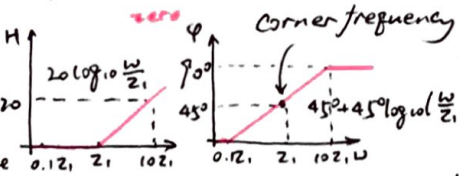
③ simple pole/zero

$(1+j\omega/z_1)$: $H_{dB} = 20\log_{10}|1+\frac{j\omega}{z_1}|$

zero $\quad \varphi = \arctan\frac{\omega}{z_1}$

$1/(1+j\omega/p_1)$: slope of H: -20dB/decade

pole $\quad$ slope of φ: -45°dB/decade

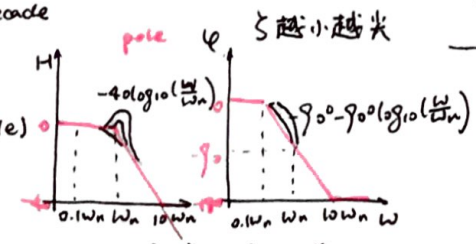


④ quadratic pole/zero

$1/[1+j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]$: (pole)

$H_{dB} = -20\log_{10}|\cdots|$

$\varphi = -\arctan\left(\frac{2\zeta_2\omega/\omega_n}{1-\omega^2/\omega_n^2}\right)$



zero is similar except

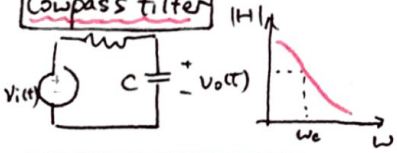
40dB/decade $\quad$ 90°/decade

Passive Filters

$H_{(\omega)} = \frac{V_o}{V_i} = \frac{1}{1+j\omega RC}$

$|H(\omega_c)| = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = \frac{1}{RC}$

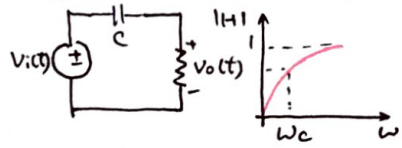
lowpass filter



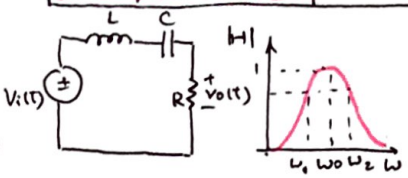
highpass filter

$H(\omega) = \frac{V_o}{V_i} = \frac{j\omega RC}{1+j\omega RC}$

$\omega_c = \frac{1}{RC}$



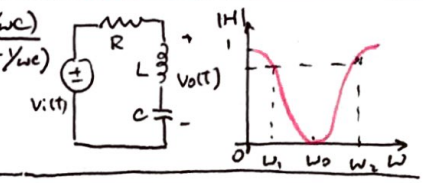
bandpass filter



$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R+j(\omega L - 1/\omega C)}$

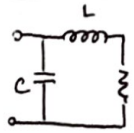
$\omega_0 = \frac{1}{\sqrt{LC}}$

bandstop filter

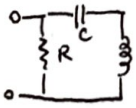


$H(\omega) = \frac{V_o}{V_i} = \frac{j(\omega L - 1/\omega C)}{R+j(\omega L - 1/\omega C)}$

$\omega_0 = \frac{1}{\sqrt{LC}}$



$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$



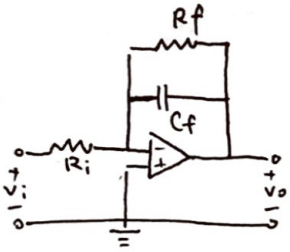
$\omega_0 = \frac{1}{\sqrt{LC}}$

Active Filters

first order low-pass filter

$H(\omega) = -\frac{R_f}{R_i}\frac{1}{1+j\omega/\omega_c}$

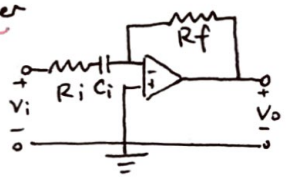
$\omega_c = 1/(R_f C_f)$



first order high-pass filter

$H(\omega) = -\frac{R_f}{R_i}\frac{j\omega/\omega_c}{1+j\omega/\omega_c}$

$\omega_c = 1/(R_i C_i)$

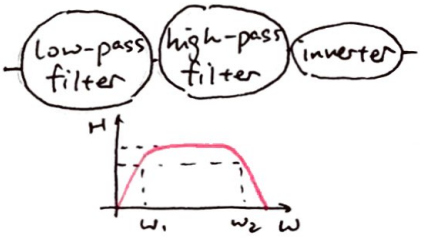


band-pass filter

H 相乘

$\omega_1 = 1/(RC_2)$ $\omega_2 = 1/(RC_1)$

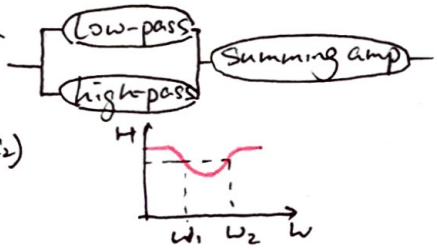
$\omega_0 = \sqrt{\omega_1\omega_2}$ $BW = \omega_2 - \omega_1$



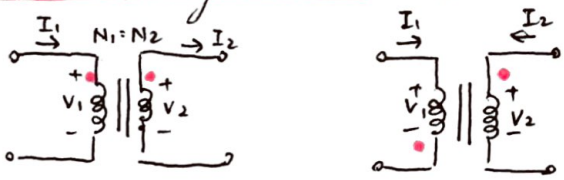
band-reject filter

H 相加

$\omega_1 = 1/(RC_1)$ $\omega_2 = 1/(RC_2)$



Ideal Transformers (k=0)



$\frac{V_2}{V_1} = \frac{N_2}{N_1}$ $\frac{I_2}{I_1} = \frac{N_1}{N_2}$ $\frac{V_2}{V_1} = -\frac{N_2}{N_1}$ $\frac{I_2}{I_1} = \frac{N_1}{N_2}$

$Z_{in} = \frac{1}{n^2} Z_L$ (reflected)

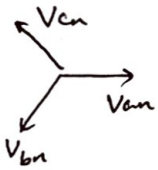
$V_{in} = \frac{V}{n}$

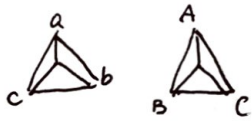
$I_{in} = nI$

§12 Three-Phase Circuit

$$\begin{cases} V_{an} = V_p\angle 0^\circ \\ V_{bn} = V_p\angle -120^\circ \\ V_{cn} = V_p\angle 120^\circ \end{cases}$$

positive





Y-Y $V_L = \sqrt{3}V_p$

$$\begin{cases} \tilde{V}_{ab} = \tilde{V}_{an}\angle 30^\circ \\ \tilde{V}_{bc} = \tilde{V}_{bn}\angle 30^\circ \\ \tilde{V}_{ca} = \tilde{V}_{cn}\angle 30^\circ \end{cases}$$

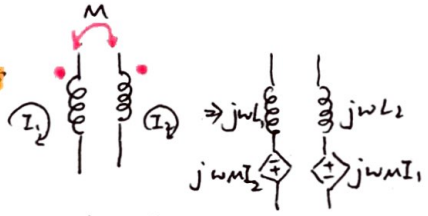
Y-Δ $V_L = \sqrt{3}V_p$

$$\begin{cases} \tilde{V}_{ab} = \tilde{V}_{an}\angle 30^\circ \\ \tilde{V}_{bc} = \tilde{V}_{bc}\angle 30^\circ \\ \tilde{V}_{ca} = \tilde{V}_{cn}\angle 30^\circ \end{cases}$$

$I_L = \sqrt{3}I_p$

$\tilde{I}_a = \tilde{I}_{AB}\sqrt{3}\angle -30^\circ$

§13 Magnetically Coupled Circuits



Self-inductance $v = L\frac{di}{dt}$

Mutual-inductance

in series $\begin{cases} L = L_1 + L_2 + 2M & \text{(aiding)} \\ L = L_1 + L_2 - 2M \end{cases}$

in parallel $L = \frac{L_1 + L_2 - M^2}{L_1 + L_2 - 2M}$ (opposing)

energy in a coupled circuit $w = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 + MI_1I_2$ (同进)

$k = \frac{M}{\sqrt{L_1L_2}}$ $(0 \le k \le 1)$

$w = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 - MI_1I_2$ (一进一出)

Δ-Δ $I_L = \sqrt{3}I_p$

$\tilde{I}_a = \tilde{I}_{AB}\sqrt{3}\angle -30^\circ$

Δ-Y

line > voltage
phase

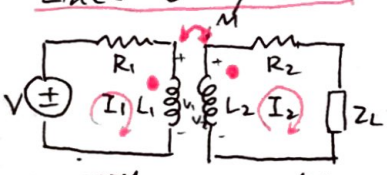
line > current
phase

$p(t) = 3V_pI_p\cos(\theta) = \text{const}$

$P_p = V_pI_p\cos\theta$ $Q_p = V_pI_p\sin\theta$

$S_p = P_p + jQ_p = \tilde{V}_p\tilde{I}_p^*$

Linear transformer



primary winding

secondary winding

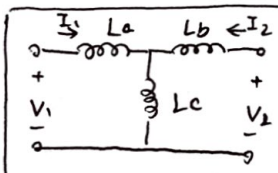
$$\begin{cases} V_1 = L_1\frac{di_1}{dt} - M\frac{di_2}{dt} \\ V_2 = M\frac{di_1}{dt} - L_2\frac{di_2}{dt} \end{cases}$$

$$\Rightarrow \begin{cases} \tilde{V}_1 = j\omega L_1\tilde{I}_1 - j\omega M\tilde{I}_2 \\ \tilde{V}_2 = j\omega M\tilde{I}_1 - j\omega L_2\tilde{I}_2 \end{cases}$$

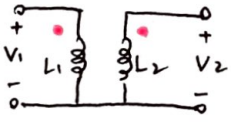
$Z_{in} = \frac{\tilde{V}_1}{\tilde{I}_1} = R_1 + j\omega L_1 + \frac{\omega^2M^2}{R_2 + j\omega L_2 + Z_L}$

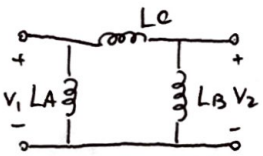
reflected impedience

(不受点影响)



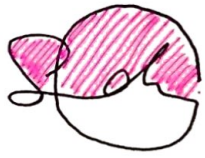
$$\begin{cases} L_a = L_1 - M \\ L_b = L_2 - M \\ L_c = L_3 - M \end{cases}$$





$$\begin{cases} L_A = \frac{L_1L_2 - M^2}{L_2 - M} \\ L_B = \frac{L_1L_2 - M^2}{L_1 - M} \\ L_C = \frac{L_1L_2 - M^2}{M} \end{cases}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$



VE215 Final